# Financial Engineering Lab (MA374)

Lab 10 Report: Option Pricing and Volatility Surface Analysis

Your Name / Roll Number

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## Contents

# 1 Introduction and Data Collection (Question 1)

As per the assignment requirements, option data comprising closing prices of calls and puts across various maturities and strike prices were collected for the NIFTY index and select individual stocks (e.g., INFY, RELIANCE, ITC). The stocks were chosen to represent different sectors and correspond to the historical database `nsedata1`.

The collected data is structured with columns including Date, Expiry, Strike Price, Option Type, Close (Option Price), Open Interest, and Underlying Value. The time to maturity $T-t$ (in years) is calculated assuming a 365-day calendar year, and the risk-free interest rate is assumed to be $r = 5\%$.

# 2 Option Data Analysis (Question 2)

## 2.1 Part A: Option Price Analysis

The behavior of option prices across different strikes and maturities is analyzed by visualizing the data in both three dimensions and two dimensions.

### 2.1.1 Graphical Representation

Below are the visualizations for the NIFTY index options. Similar plots were generated for individual stocks and are available in the `Output_Plots` directory.

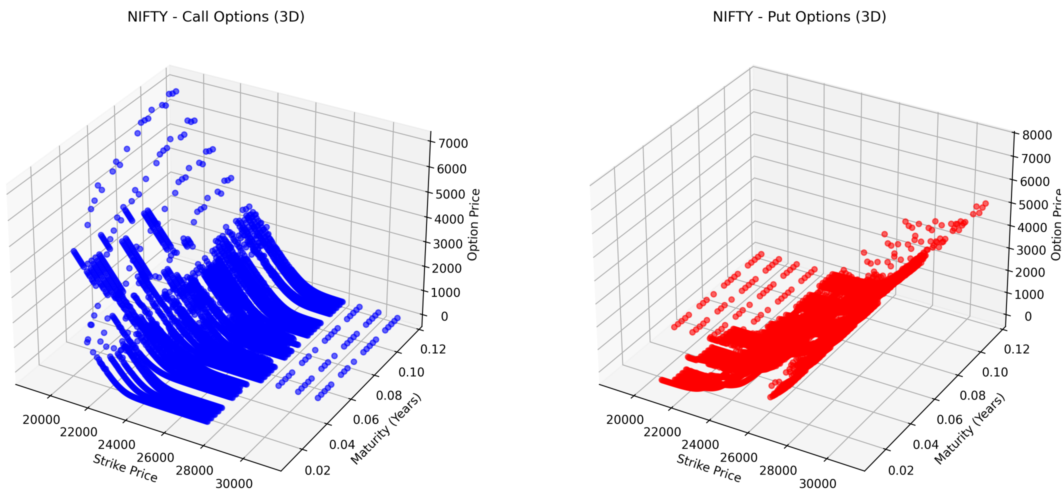


Figure 1: 3D Plot of NIFTY Call and Put Option Prices

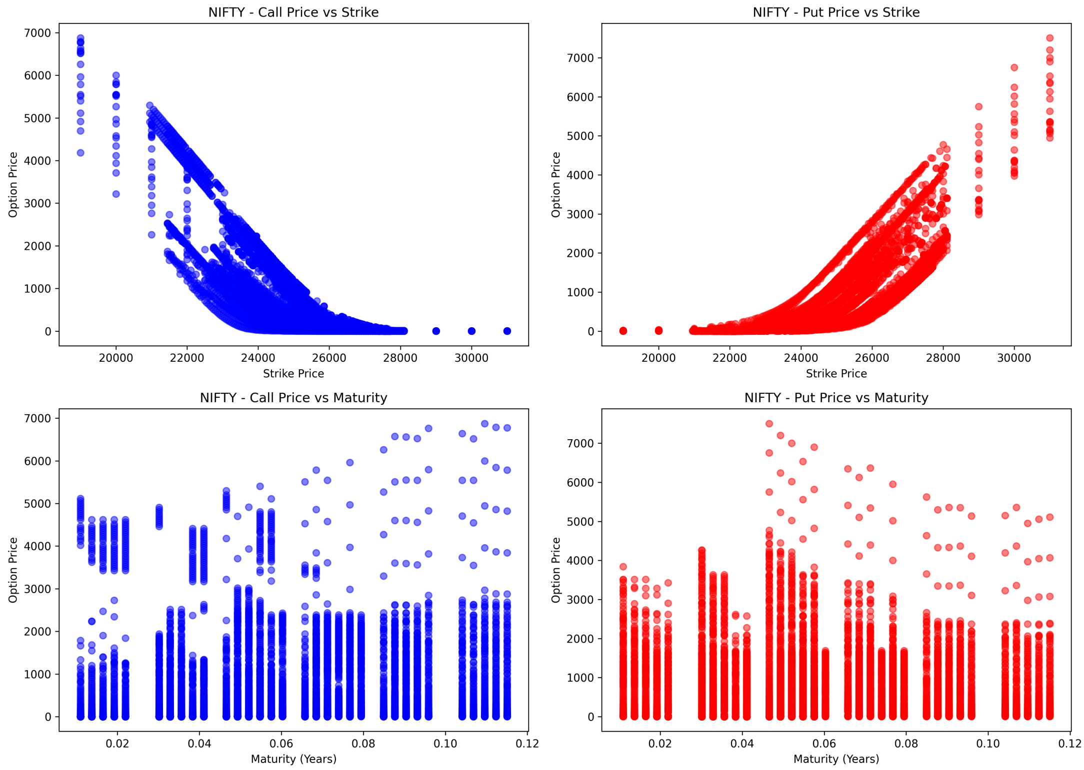


Figure 2: 2D Projections: Option Price vs. Strike and Option Price vs. Maturity

### 2.1.2 Observations

Examining the two-dimensional plots (Figure 2) yields the following theoretical verifications:

- **Option Price vs. Strike Price:** The call option prices strictly decrease as the strike price increases, reflecting the lower probability of the option expiring in-the-money. Conversely, put option prices exhibit an upward-sloping curve, as the right to sell at a higher strike price inherently carries more value.

- **Option Price vs. Maturity:** For a fixed strike price, both call and put options generally increase in value as the time to maturity increases. This reflects the "time value" of options; a longer horizon provides a higher probability for the underlying asset to move favorably before expiration.

## 2.2 Part B: Implied Volatility (IV)

The Implied Volatility ($\sigma_{IV}$) is extracted by equating the market price of the option to the Black-Scholes-Merton (BSM) pricing formula and solving for volatility using the Newton-Raphson root-finding method.

The iterative update rule used is:

$$\sigma_{n+1} = \sigma_n - \frac{C_{BSM}(\sigma_n) - C_{Market}}{\mathcal{V}(\sigma_n)} \tag{1}$$

where $\mathcal{V}(\sigma_n)$ is the option Vega, defined as $\frac{\partial C}{\partial \sigma}$.

### 2.2.1 Graphical Representation

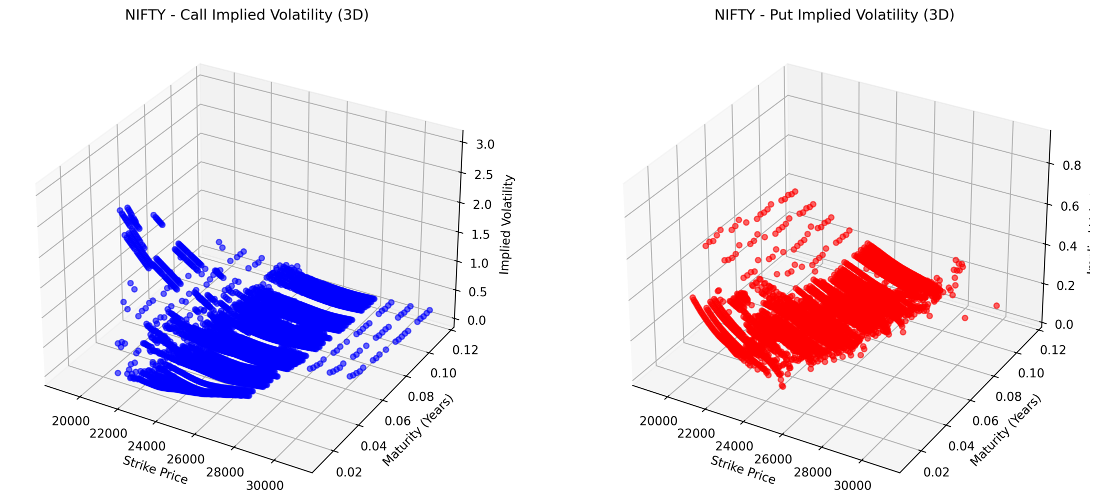


Figure 3: 3D Volatility Surface for NIFTY Options

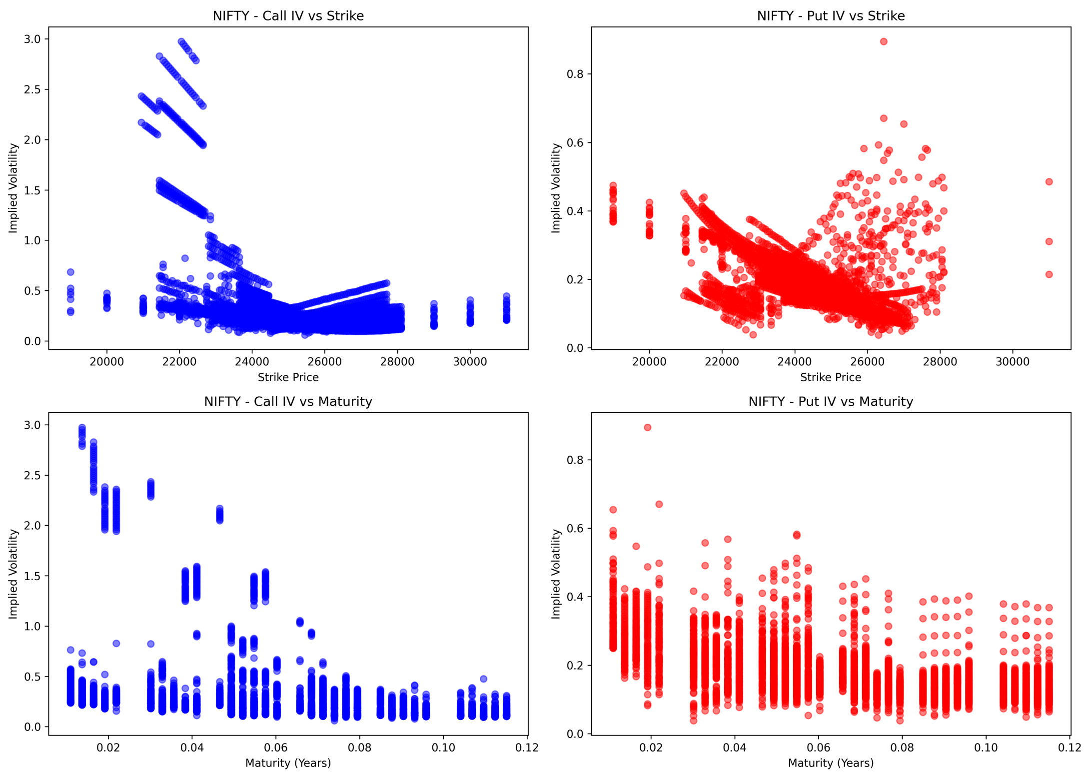


Figure 4: 2D Projections: Implied Volatility vs. Strike and Maturity

### 2.2.2 Observations

- **Implied Volatility vs. Strike (The Volatility Smirk/Smile):** Unlike the BSM assumption of constant volatility, the empirical data shows a distinct skew or "smile". Deep out-of-the-money (OTM) puts trade at significantly higher implied volatilities than at-the-money (ATM) options. This skew exists because market participants are willing to pay a premium for downside protection against market crashes.

- **Implied Volatility vs. Maturity (Term Structure):** The plots across different maturities reveal the term structure of volatility. It illustrates the market's expectation of how volatility will behave over time, often showing mean-reverting properties where short-term shocks smooth out over longer horizons.

## 2.3 Part C: Historical vs. Implied Volatility

Historical Volatility (HV) is computed by utilizing the historical daily log returns of the underlying asset. For an option with a maturity of $T$, the historical dataset traces back a corresponding period of $T$ from $t = 0$. The standard deviation of these returns is then annualized.

### 2.3.1 Tabular Results

The computed Historical Volatilities alongside their respective Implied Volatilities have been exported to CSV files (`NIFTY_Volatility_Comparison.csv` and `Stock_Volatility_Comparison.c`
Due to the large volume of observations, the data is attached electronically, and a graphical summary is provided below.

### 2.3.2 Graphical Comparison

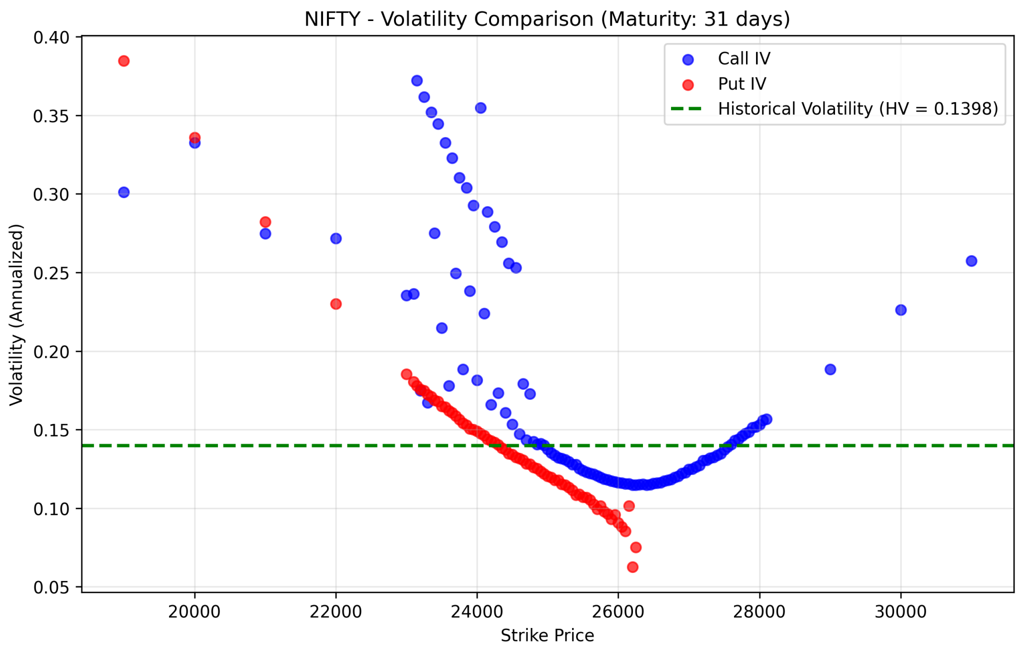


Figure 5: Implied Volatility (Smile) vs. Historical Volatility (Baseline)

### 2.3.3 Observations and Comparison

Comparing the two volatilities highlights a fundamental pricing dynamic:

- **Static vs. Dynamic:** Historical Volatility serves as a flat, backward-looking baseline for a specific maturity period. Implied Volatility is forward-looking and varies dynamically across different strike prices.
- **Volatility Risk Premium:** In almost all standard market conditions, the Implied Volatility is generally higher than the Historical Volatility ($IV > HV$). This gap represents the *Volatility Risk Premium*. Option sellers demand higher compensation than what historical price movements suggest to account for the asymmetric, unlimited tail risk they bear.